Arjun Desai

Software Engineer Intern

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EXPERIENCE

Stripe | Jun–Aug 2023
Software Engineer Intern
- Built payment processing microservices in Go reducing transaction latency by 47ms (18% improvement) for 50K+ daily requests; code merged to production within 2 days
- Designed and implemented comprehensive unit/integration test suite increasing coverage from 62% to 89%, preventing 3 critical bugs in staging
- Collaborated with 6-person platform team to migrate legacy payment reconciliation system to Kafka, reducing data sync time by 2.3 hours (from 5.2h to 2.9h per day)

Cloudflare | Jan–May 2023
Full-Stack Engineer Intern
- Developed React + TypeScript dashboard UI for DDoS analytics, consumed by 8K+ paying customers; improved page load time from 3.2s to 840ms (74% reduction)
- Optimized PostgreSQL queries using explain analyze and indexing strategy, reducing API response time by 65% for user onboarding endpoints serving 2M+ monthly active users
- Maintained 2 open-source projects with 4.2K+ GitHub stars; reviewed 12 community PRs and merged critical bugfixes, boosting release download rate by 23%

Datadog | Jul–Sep 2022
Software Engineer Intern
- Built real-time monitoring alerts backend in Python + FastAPI processing 500K+ metrics per second; system achieved 99.97% uptime SLA across 3 availability zones
- Engineered efficient time-series data aggregation reducing memory footprint by 52% (from 2.8GB to 1.3GB per worker node) without compromising query latency
- Designed schema migrations framework supporting 0-downtime deployments across 45K+ customer databases; reduced deployment window from 6 hours to 12 minutes

Wise | Sep 2024–Present
Graduate Software Engineer
- Architected end-to-end payments orchestration service in Java + Spring Boot handling 200K+ daily international transfers; improved transaction success rate from 94.2% to 98.1%
- Implemented distributed tracing instrumentation across 12 microservices using OpenTelemetry, reducing mean time to resolution (MTTR) for production issues by 31 minutes
- Deployed real-time FX rate caching layer (Redis) decreasing quote latency by 340ms (92% improvement) and reducing database load by 67% across 15 backend instances

PROJECTS

CacheFlow | 2024
TypeScript, React, Node.js, Redis, PostgreSQL
- Built open-source distributed caching framework with automatic invalidation; achieved 8.2K GitHub stars and 150K+ monthly npm downloads within 6 months of launch
- Implemented adaptive TTL mechanism and compression algorithms reducing memory usage by 58% compared to standard Redis implementations; benchmarks published in blog post (2.3K reads)

GraphQL Schema Validator | 2023
Rust, GraphQL, CLI
- Developed compile-time schema validation tool for GraphQL APIs; 6.5K GitHub stars with 89K monthly downloads; used internally by 50+ engineering teams at 8 Fortune 500 companies
- Achieved 40× performance improvement over Node.js-based competitors through Rust implementation; validation time: 120ms vs. 4.8s for enterprise-scale schemas (2K+ types)

StreamSync | 2023–2024
Go, gRPC, Protobuf, Kubernetes
- Created real-time data synchronization library for microservices; supports 500K+ events/second throughput with sub-50ms latency; 3.1K GitHub stars and active maintenance with 14 contributors
- Implemented consensus-based conflict resolution achieving 99.99% eventual consistency guarantees; published technical deep-dive (1.8K medium.com claps)

PerformanceInsight | 2022–2023
Python, FastAPI, Vue.js, ClickHouse
- Built full-stack web application for application performance monitoring; supports 4M+ metric ingestion per minute with <500ms query response time across 90-day retention
- Designed novel columnar data compression technique reducing storage costs by 44% (from £8K/month to £4.5K/month) without sacrificing query performance; presented at PyCon 2023 (320 attendees)

EDUCATION

Imperial College London | 2020–2024
MEng Computing (First Class Honours)
- Achieved 3.92/4.0 GPA with Distinction in final year; awarded Imperial Excellence Scholarship (top 2% of cohort)
- Led Department Computing Society (150+ members) as President, organized weekly technical workshops and secured £12K sponsorship from industry partners

University of Cambridge | 2023
Cambridge Summer Researcher (CS Program)
- Completed 8-week research internship on distributed systems optimization; co-authored paper on consensus algorithms achieving 34% latency reduction
- Mentored 4 junior developers on systems design patterns; mentor feedback scores averaged 4.8/5.0

EXTRACURRICULAR

Python Software Foundation | 2022–Present

Open Source Maintainer

- Maintain 2 core libraries with 180K+ weekly PyPI downloads; triaged and resolved 240+ issues and merged 95 community PRs from 45+ contributors worldwide
- Authored 3 technical documentation guides and 2 architectural RFCs; community feedback: 4.9/5.0 average helpfulness rating across 1.2K responses

European Software Engineering Conference (ESEC) | 2023–2024

Speaker & Workshop Facilitator

- Delivered 2 conference talks on 'Scaling Microservices to 1M+ RPS' (ESEC 2024, 380 attendees); session feedback: 4.7/5.0; published accompanying 22-page technical report with 8.3K downloads
- Facilitated full-day workshop on 'Production-Grade Go for Backend Engineers' at 3 major conferences; trained 180+ engineers; post-workshop survey: 94% would recommend

SKILLS

Languages: Python, JavaScript/TypeScript, Go, Java, Rust, SQL

Technologies: React, Node.js, FastAPI, Spring Boot, GraphQL, gRPC, PostgreSQL, Redis, Kafka, Kubernetes, Docker, AWS (S3, Lambda, RDS, ECS)

Tools: Git, GitHub, GitLab, Jira, Datadog, Prometheus, PagerDuty, Figma, VS Code, IntelliJ IDEA